Name			
USN			
Date	Conduction	Record	Viva
Allotted Marks			
Obtained Marks			

EXPERIMENT 3

Develop a Map Reduce program that mines weather data and displays appropriate messages indicating the weather conditions of the day

Aim: To develop a Map Reduce program that mines weather data and displays appropriate messages indicating the weather conditions of the day.

Theory:

Weather data is typically huge (hourly readings from thousands of sensors). MapReduce allows efficient, scalable processing of such data to extract

- Maximum or minimum temperatures
- Rainfall status
- Wind speeds
- Condition messages (e.g., “Hot Day,” “Rainy Day”)

The Mapper will:

- Read each line of weather data.
- Extract the date, temperature, humidity, and rainfall.
- Output the date as the **key** and weather metrics as the **value**.

The Reducer will:

- Receive all weather records grouped by date.
- Compute maximum temperature, average humidity, total rainfall.
- Based on these, generate a descriptive weather message.

Procedure:

Results: